A First Analysis of Feedback 1

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Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

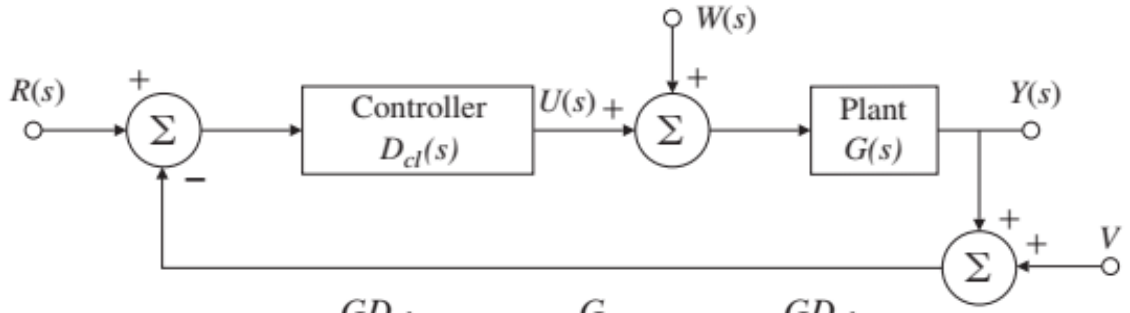
Content

1. The basic equations of control
2. Control steady-state error to polynomial inputs: system type

1. The basic equations of control

The 2 basic structures for control:

Closed-loop



$$Y_{cl} = \frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V,$$

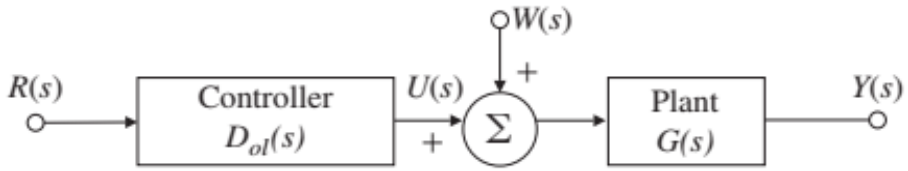
$$U = \frac{D_{cl}}{1+GD_{cl}}R - \frac{GD_{cl}}{1+GD_{cl}}W - \frac{D_{cl}}{1+GD_{cl}}V.$$

$$E_{cl} = R - Y_{cl}.$$

$$E_{cl} = R - \left[\frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V\right]$$

$$= \frac{1}{1+GD_{cl}}R - \frac{G}{1+GD_{cl}}W + \frac{GD_{cl}}{1+GD_{cl}}V.$$

Open-loop



$$Y_{ol} = GD_{ol}R + GW.$$

$$E_{ol} = R - Y_{ol},$$

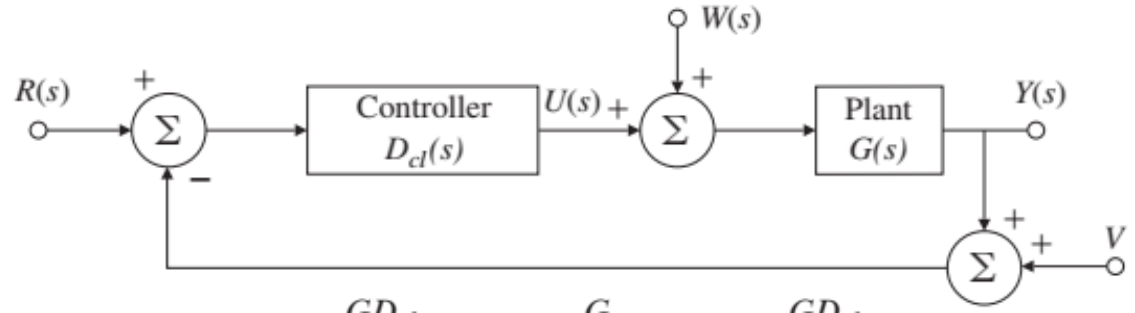
$$= R - [GD_{ol}R + GW]$$

$$= [1 - GD_{ol}]R - GW.$$

1. The basic equations of control

The 2 basic structures for control:

Closed-loop



$$Y_{cl} = \frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V,$$

$$U = \frac{D_{cl}}{1+GD_{cl}}R - \frac{GD_{cl}}{1+GD_{cl}}W - \frac{D_{cl}}{1+GD_{cl}}V.$$

$$E_{cl} = R - Y_{cl}.$$

$$E_{cl} = R - \left[\frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V\right]$$

$$= \frac{1}{1+GD_{cl}}R - \frac{G}{1+GD_{cl}}W + \frac{GD_{cl}}{1+GD_{cl}}V.$$

Re-writing the equations in a compact form

$$Y_{cl} = \mathcal{T}R + G\mathcal{S}W - \mathcal{T}V,$$

$$U = D_{cl}\mathcal{S}R - \mathcal{T}W - D_{cl}\mathcal{S}V,$$

$$E_{cl} = \mathcal{S}R - G\mathcal{S}W + \mathcal{T}V,$$

$$\mathcal{S} = \frac{1}{1+GD_{cl}};$$

$$\mathcal{T} = \frac{GD_{cl}}{1+GD_{cl}}$$

1. The basic equations of control

With these equations, we will explore the four basic objectives for both the open-loop and the closed-loop cases:

1. Stability
2. Tracking
3. Regulation
4. Sensitivity

1. The basic equations of control

1. Stability: open-loop controller

- An open-loop structure **cannot** be used if the plant is unstable
- An open-loop structure **cannot** be used to make an unstable plant to be stable
- It is not possible to **exactly** cancel a RHP pole of a(s) (Plant) with a zero of c(s) (Controller)
  - The **slightest noise or disturbance** will cause the output to **grow** until stopped by **saturation** or system **failure**
- It is not possible to **exactly** cancel a RHP zero of b(s) (Plant) with a pole of d(s) (Controller)

$$G(s) = \frac{b(s)}{a(s)}$$

$$D_{ol}(s) = \frac{c(s)}{d(s)}$$

$$GD_{ol} = \frac{bc}{ad}$$

1. The basic equations of control

1. Stability: closed-loop controller

- The characteristic equation provides **more freedom** to the controller design
- It is necessary to avoid cancellations in the RHP (unstable cancelations)
- It is possible to adjust the parameters of the controller to **force all the roots** of the characteristic equation in the **LHP**

$$1 + GD_{cl} = 0,$$

$$1 + \frac{b(s)c(s)}{a(s)d(s)} = 0,$$

$$a(s)d(s) + b(s)c(s) = 0.$$

1. The basic equations of control

1. Stability: closed-loop controller

Example: express the controller's parameters K and δ in terms of ζ and ωn in the close loop system formed by the plant G(s) (inverted pendulum) and the controller Dcl(s) lead-lag. Assume γ = 1.

$$G(s) = \frac{1}{s^2-1}$$

$$D_{cl}(s) = \frac{K(s+\gamma)}{s+\delta}$$

1. The basic equations of control

1. **Stability: closed-loop controller**

Example: express the controller's parameters K and δ in terms of ζ and ωn in the close loop system formed by the plant G(s) (inverted pendulum) and the controller Dcl(s) lead-lag. Assume γ = 1.

$$G(s) = \frac{1}{s^2-1} \qquad D_{cl}(s) = \frac{K(s+\gamma)}{s+\delta}$$

$$(s+1)(s-1)(s+\delta) + K(s+\gamma) = 0$$

$$\delta = 2\zeta w_n + 1 \qquad K = w_n^2 + 2\zeta w_n + 1$$

For ζ = 0.707 and wn = 2 rad/s
δ = 3.8280
K = 7.8280

1. The basic equations of control

2. Tracking

- The tracking problem is to cause the **output** to **follow** the **reference** input as closely as possible

Open-loop

- The controller Dcl(s) can “cancel” the plant G(s) and impose the desired transfer function
  - G(s) must be stable
  - The transfer function of the must be proper (num(zeros) ≤ num(poles))
  - The imposed dynamic cannot be too fast (regarding the plant dynamic)
  - The poles of G(s) cannot be close to the origin => G(s) can become unstable if there is a small variation in its parameters

1. The basic equations of control

2. Tracking

- **Exercise.** For a plant having the transfer function G(s) = $1/(s^2+3s+9)$ it is proposed to use a controller in a unity feedback system and having the transfer function Dcl(s) = $(c_2s^2+c_1s+c_0)/(s(s+d_1))$. Solve for the parameters of this controller so the closed loop will have the characteristic equation $(s + 6)(s + 3)(s^2 + 3s + 9) = 0$

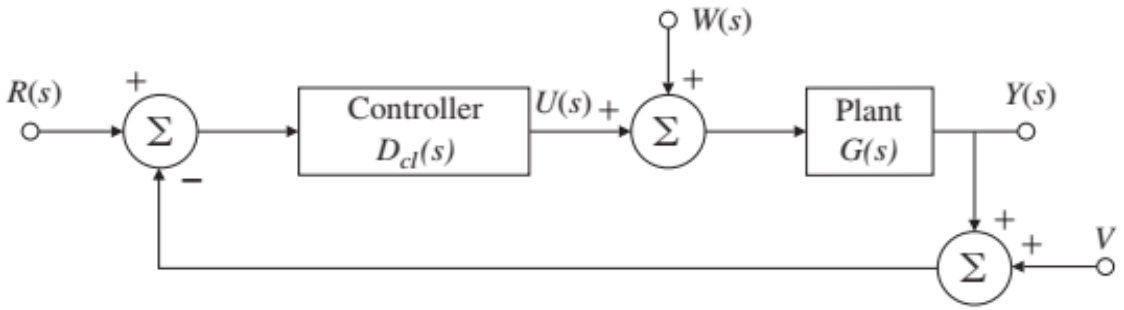
{Answer: $c_2 = 18$, $c_1 = 54$, $c_0 = 162$, $d_1 = 9$}

This procedure is named **pole-placement**

1. The basic equations of control

3. Regulation

- The problem of regulation is to **keep the error small** when the **reference is a constant** setpoint and **disturbances** are present
  - **Open-loop** controller is **useless** for regulation
  - **Closed-loop** controller is **useful** for regulation
- In a closed-loop controller there is a conflict between W and V for a good controller



$$E_{cl} = R - \left[\frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V\right]$$

$$= \frac{1}{1+GD_{cl}}R - \frac{G}{1+GD_{cl}}W + \frac{GD_{cl}}{1+GD_{cl}}V.$$

Increment of Dcl:

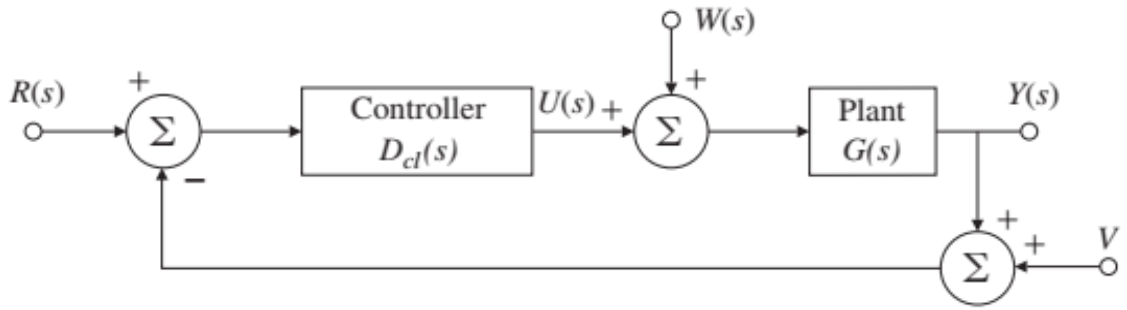
- Reduces effect of W on Ecl
- Increases effect of V on Ecl

1. The basic equations of control

3. Regulation

Solution to the conflict:

- W and V depend on frequency
  - W usually high for low frequency (common case W is a bias => high for f = 0)
  - V usually high for high frequencies (common case high noise for high frequencies)
- Dcl(s) is designed to be
  - High for low frequencies => reduce effect of W on Ecl(s)
  - Low for high frequencies => reduce effect of V on Ecl(s)



$$E_{cl} = R - \left[\frac{GD_{cl}}{1+GD_{cl}}R + \frac{G}{1+GD_{cl}}W - \frac{GD_{cl}}{1+GD_{cl}}V\right]$$

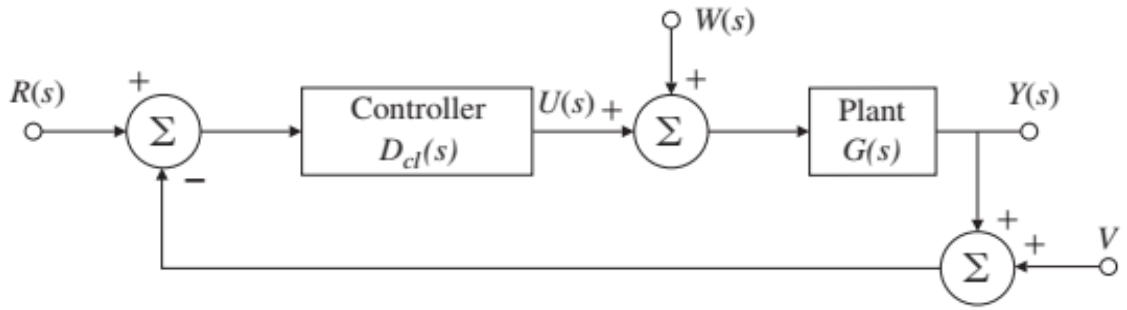
$$= \frac{1}{1+GD_{cl}}R - \frac{G}{1+GD_{cl}}W + \frac{GD_{cl}}{1+GD_{cl}}V.$$

1. The basic equations of control

3. Regulation

Exercise.

- Show that if W is a constant bias and if Dcl(s) has a pole at s = 0, then the error due to this bias will be zero.
- However, show that if G(s) has a pole at zero, the error due to this bias will not be zero.



1. The basic equations of control

3. Sensitivity

For variations in the gain of the plant G the sensitivity is defined as S_G^T where **T is the transfer function** of the controlled system and **G is the gain** of the plant

$$\mathcal{S}_G^T = \frac{\frac{\delta T_{ol}}{T_{ol}}}{\frac{\delta G}{G}}$$

1. The basic equations of control

3. Sensitivity

- For open-loop
  - Sensitivity is 1
  - 10% of variation in G produces 10% of variation in T_{ol}

$$T_{ol} + \delta T_{ol} = D_{ol}(G + \delta G) = D_{ol}G + D_{ol}\delta G = T_{ol} + D_{ol}\delta G.$$

$$\delta T_{ol} = D_{ol}\delta G. \qquad T_{ol} = D_{ol}G$$

$$\mathcal{S}_G^T = \frac{\frac{\delta T_{ol}}{T_{ol}}}{\frac{\delta G}{G}},$$

$$= \frac{G}{T_{ol}}\frac{\delta T_{ol}}{\delta G}$$

$$S_G^T = 1$$

1. The basic equations of control

3. Sensitivity

- For closed-loop (unitary feedback)

Calculating δTcl from the derivative

$$\delta T_{cl} = \frac{dT_{cl}}{dG}\delta G.$$

$$T_{cl} + \delta T_{cl} = \frac{(G + \delta G)D_{cl}}{1 + (G + \delta G)D_{cl}}$$

$$T_{cl} + \delta T_{cl} = \frac{(G + \delta G)D_{cl}}{1 + (G + \delta G)D_{cl}}$$

$$T_{cl} = \frac{GD_{cl}}{1 + GD_{cl}}$$

$$\mathcal{S}_G^{T_{cl}} \triangleq \text{sensitivity of } T_{cl} \text{ with respect to } G,$$

$$\mathcal{S}_G^{T_{cl}} \triangleq \frac{G}{T_{cl}}\frac{dT_{cl}}{dG},$$

$$\mathcal{S}_G^{T_{cl}} = \frac{G}{GD_{cl}/(1 + GD_{cl})}\frac{(1 + GD_{cl})D_{cl} - D_{cl}(GD_{cl})}{(1 + GD_{cl})^2}$$

$$= \frac{1}{1 + GD_{cl}}.$$

1. The basic equations of control

3. Sensitivity

- For closed-loop (unitary feedback)
  - 1 + GDcl > 1, then the sensitivity < 1
  - The sensitivity of a closed-loop system is less than the one of an open-loop system by a factor of 1/(1 + GDcl)
  - This sensitivity depends on the frequency
  - This sensitivity is usually named just S

$$\mathcal{S}_G^{T_{cl}} = \frac{G}{GD_{cl}/(1+GD_{cl})} \frac{(1+GD_{cl})D_{cl} - D_{cl}(GD_{cl})}{(1+GD_{cl})^2}$$
$$= \frac{1}{1+GD_{cl}}.$$

$$\mathcal{S} \triangleq \frac{1}{1+GD_{cl}}$$

The transfer function of the closed-loop system can be seen as the **complementary sensitivity function**

$$\mathcal{T} \triangleq \frac{GD_{cl}}{1+GD_{cl}} = 1 - \mathcal{S}$$

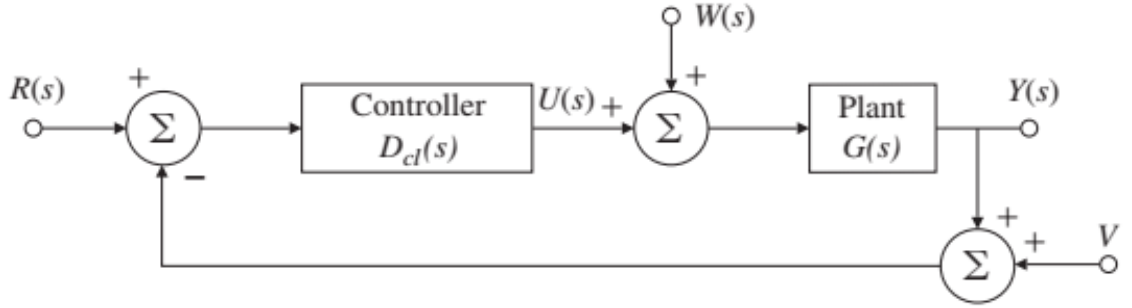
2. Control steady-state error to polynomial inputs: system type

- There are systems that require references (R(s)) to change with time with a given form:
  - Step
  - Ramp
  - Parabola
- Examples:
  - Speed reference in a motor must change as ramps (no step)
  - Speed reference in an elevator must change as a ramp (when accelerates and when deaccelerate
  - Antena tracking the position of a satellite (parabola variation of the reference angle)

2. Control steady-state error to polynomial inputs: system type

This analysis is performed for a system:

- In closed-loop with unity feedback
- Stable (otherwise no sense)
- In steady-state (after the transient)
- Polynomial input



W = V = 0 $$E = \frac{1}{1 + GD_{cl}} R = \mathcal{S} R, \qquad \mathcal{S} = \frac{1}{1 + GD_{cl}}$$

For a polynomial input r(t) $$r(t) = t^k / k! 1(t) \qquad R = \frac{1}{s^{k+1}}$$

Using the following names (from mechanical system)
k = 0: “position” inputs (**steps**)
k = 1: “velocity” inputs (**ramps**)
k = 2: “acceleration” inputs (**parabolas**)

2. Control steady-state error to polynomial inputs: system type

Applying the final value theorem

$$\lim_{t\to\infty} e(t) = e_{ss} = \lim_{s\to 0} sE(s),$$
$$= \lim_{s\to 0} s\frac{1}{1+GD_{cl}}R(s),$$
$$= \lim_{s\to 0} s\frac{1}{1+GD_{cl}}\frac{1}{s^{k+1}}$$

Applying the final value theorem with:

- **GDcl with 0 poles at the origin => Type 0**
- R(s) = 1/s (unit-step reference)
- **e_{ss} different than 0**

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+GD_{cl}}\frac{1}{s},$$
$$\frac{e_{ss}}{r_{ss}} = \frac{e_{ss}}{1} = e_{ss} = \frac{1}{1+GD_{cl}(0)}$$

Where: $r_{ss} = \lim_{t\to\infty} r(t) = 1$

Position error constant: $GD_{cl}(0) \triangleq K_p$

2. Control steady-state error to polynomial inputs: system type

Applying the final value theorem

$$\lim_{t\to\infty} e(t) = e_{ss} = \lim_{s\to 0} sE(s),$$

$$= \lim_{s\to 0} s\frac{1}{1+GD_{cl}}R(s),$$

$$= \lim_{s\to 0} s\frac{1}{1+GD_{cl}}\frac{1}{s^{k+1}}$$

Writing GDcl(s) as follows (the integrators apart)

- **0 intrators => n = 0 (type 0)**
- **1 intrator => n = 1 (type 1)**
- **Etc.**

$$GD_{cl}(s) = \frac{GD_{clo}(s)}{s^n}$$

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+\dfrac{GD_{clo}(s)}{s^n}}\frac{1}{s^{k+1}}$$

$$= \lim_{s\to 0}\frac{s^n}{s^n+K_n}\frac{1}{s^k}.$$

$$GD_{clo}(0) = K_n$$

- **n > k** then $\mathbf{e_{ss} = 0}$
- **n < k** then $\mathbf{e_{ss} = \infty}$
- **n = k = 0** then $\mathbf{e_{ss} = 1/(1 + K_0)}$
- **n = k different than 0** then $\mathbf{e_{ss} = 1/K_n}$

2. Control steady-state error to polynomial inputs: system type

$$GD_{cl}(s) = \frac{GD_{clo}(s)}{s^n}$$

$$GD_{clo}(0) = K_n$$

$$e_{ss} = \lim_{s\to 0} s \frac{1}{1 + \frac{GD_{clo}(s)}{s^n}} \frac{1}{s^{k+1}}$$

$$= \lim_{s\to 0} \frac{s^n}{s^n + K_n} \frac{1}{s^k}.$$

Constants for typical **types** from the open-loop transfer function GDcl(s)

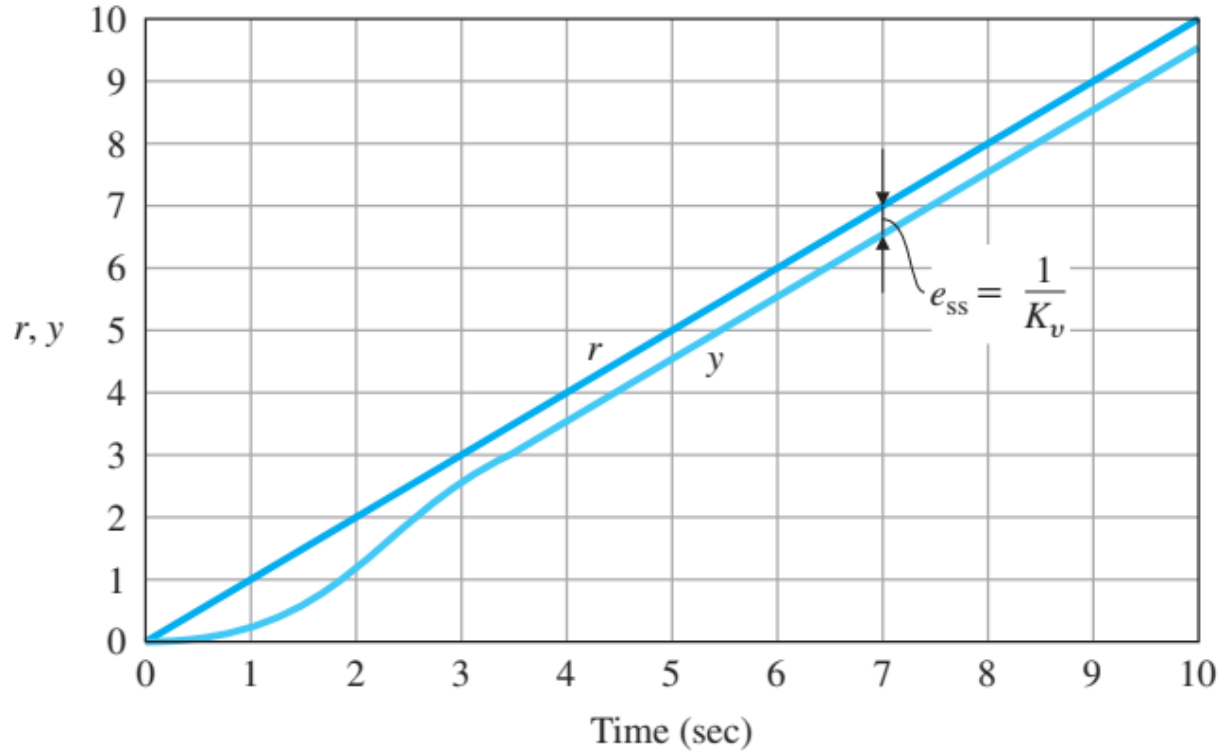
$$K_p = \lim_{s\to 0} GD_{cl}(s), \qquad n = 0$$

$$K_v = \lim_{s\to 0} sGD_{cl}(s), \qquad n = 1$$

$$K_a = \lim_{s\to 0} s^2 GD_{cl}(s), \qquad n = 2$$

2. Control steady-state error to polynomial inputs: system type

Error of a **Type 1** system for a **ramp reference**:



2. Control steady-state error to polynomial inputs: system type

Summary table:

Errors as a Function of System Type			
Type Input	**Step (position)**	**Ramp (velocity)**	**Parabola (acceleration)**
Type 0	$\frac{1}{1+K_p}$	∞	∞
Type 1	0	$\frac{1}{K_v}$	∞
Type 2	0	0	$\frac{1}{K_a}$

2. Control steady-state error to polynomial inputs: system type

Example: System Type for Speed Control

Determine the system type and the relevant error constant for speed control with proportional feedback given by $D_{cl}(s) = k_P$. The plant transfer function is $G = A/(\tau s+1)$.

2. Control steady-state error to polynomial inputs: system type

Example: System Type for Speed Control

Determine the system type and the relevant error constant for speed control with proportional feedback given by Dcl(s) = k_P. The plant transfer function is G = A/(τs+1).

$$G = \frac{A}{\tau s+1} \qquad D_{cl}(s) = k_P$$

$$GD_{cl} = \frac{k_P A}{\tau s+1}$$

- 0 poles at 0
- n = 0 (0 integrators (1/s))

$$K_p = \lim_{s\to 0} GD_{cl}(s), \qquad n = 0$$

$$K_p = k_P A$$

2. Control steady-state error to polynomial inputs: system type

Example: System Type Using Integral Control

Determine the system type and the relevant error constant for the speed control example with proportional plus integral control having controller given by

$Dcl = k_P + k_I/s.$

The plant transfer function is

$G = A/\tau s+1.$

2. Control steady-state error to polynomial inputs: system type

Example: System Type Using Integral Control

Determine the system type and the relevant error constant for the speed control example with proportional plus integral control having controller given by

Dcl = k_P + k_I/s.

The plant transfer function is

G = A/τs+1.

$$GD_{cl}(s) = \frac{A(k_P s + k_I)}{s(\tau s + 1)}$$

- 1 pole at 0
- n = 1 (1 integrators (1/s))

$$K_v = \lim_{s \to 0} sGD_{cl}(s) = Ak_I$$

2. Control steady-state error to polynomial inputs: system type

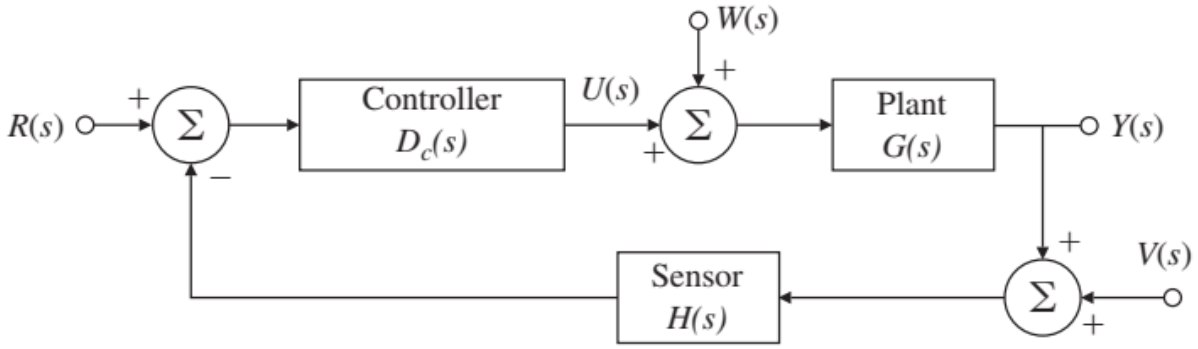
Robustness of system type

- In the **unity feedback** structure: if the process parameters change (G(s)) without removing the pole at the origin in a Type 1 system
  - **Zero steady-state error** in response to a **constant input** (R(s))
- Similar statements can be made for systems of Type 2 or higher

- **System Type** is a **robust property** with respect to **parameter changes** in the closed-loop system

2. Control steady-state error to polynomial inputs: system type

Another way to see the type of a system



$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{GD_c}{1 + GD_cH}$$

$$E(s) = R(s) - Y(s) = R(s) - \mathcal{T}(s)R(s)$$

$$\frac{E(s)}{R(s)} = 1 - \mathcal{T}(s)$$

$$E(s) = [1 - \mathcal{T}(s)]\,R(s)$$

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\,[1 - \mathcal{T}(s)]\,R(s)$$

$$E(s) = \frac{1}{s^{k+1}}\,[1 - \mathcal{T}(s)]$$

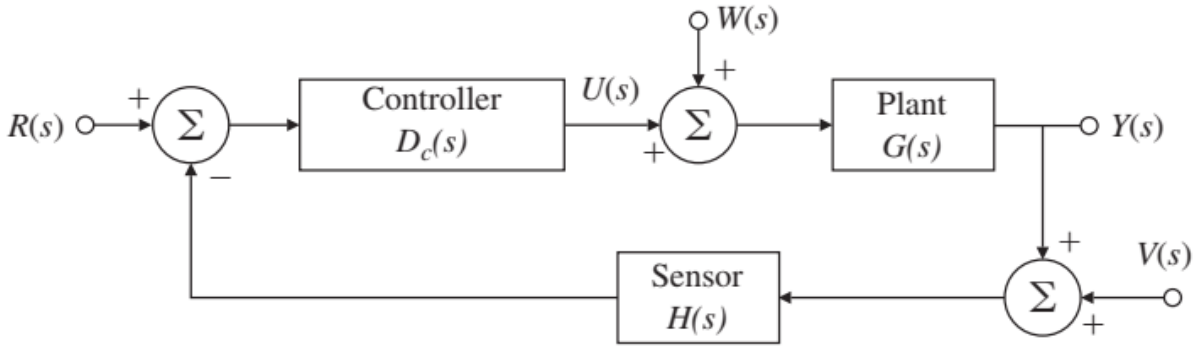
e_{ss} can be:

- 0
- Nonzero (system is type k)
- Infinite

$$e_{ss} = \lim_{s\to 0} s\frac{1 - \mathcal{T}(s)}{s^{k+1}} = \lim_{s\to 0} \frac{1 - \mathcal{T}(s)}{s^k}$$

2. Control steady-state error to polynomial inputs: system type

Example: Consider an electric motor position control problem including a nonunity feedback system caused by having a tachometer fixed to the motor shaft and its voltage (which is proportional to shaft speed) is fed back as part of the control. Determine the system type and relevant error constant with respect to reference inputs.



$$G(s) = \frac{1}{s(\tau s + 1)}$$

$$D_c(s) = k_P,$$

$$H(s) = 1 + k_t s.$$

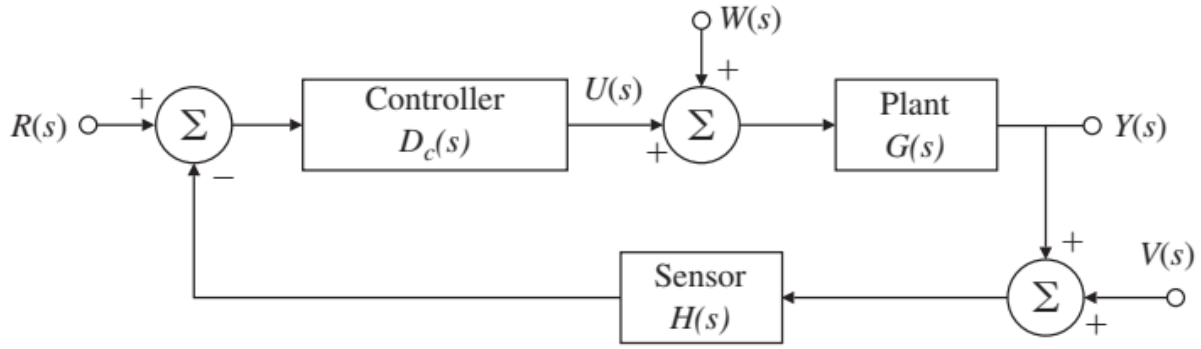
2. Control steady-state error to polynomial inputs: system type

Example:

$$G(s) = \frac{1}{s(\tau s + 1)}$$

$$D_c(s) = k_P,$$

$$H(s) = 1 + k_t s.$$



$$e_{ss} = \lim_{s \to 0} sR(s)\,[1 - \mathcal{T}(s)]$$

$$e_{ss} = \lim_{s \to 0} \frac{[1 - \mathcal{T}(s)]}{s^k} = \lim_{s \to 0} \frac{1}{s^k} \frac{s(\tau s + 1) + (1 + k_t s - 1)k_P}{s(\tau s + 1) + (1 + k_t s)k_P},$$

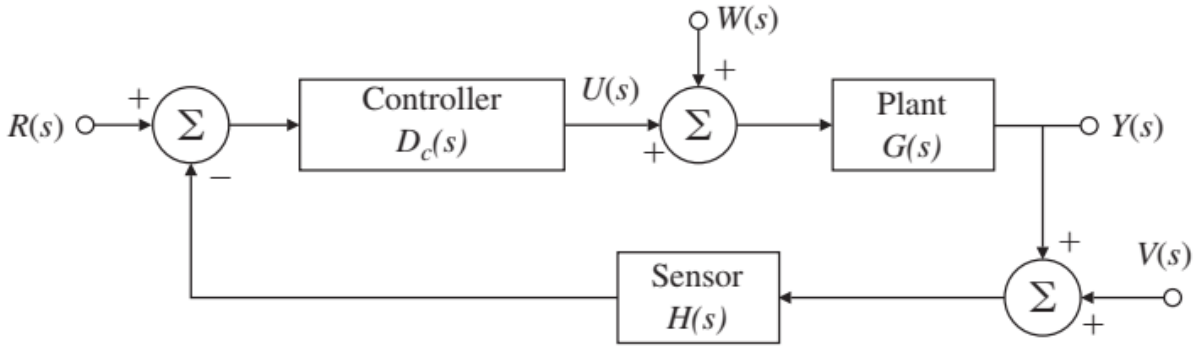
$$= 0, \qquad k = 0,$$

$$= \frac{1 + k_t k_P}{k_P}, \quad k = 1;$$

2. Control steady-state error to polynomial inputs: system type

System Type for Regulation and Disturbance Rejection

- Assuming R(s) = 0
- Regarding W(s):
  - Type 0 if e_{ss} (y_{ss}) is nonzero constant when W(s) is a step (1/s)
  - Type 1 if e_{ss} (y_{ss}) is nonzero constant when W(s) is a ramp ($1/s^2$)
  - Type 2 if e_{ss} (y_{ss}) is nonzero constant when W(s) is a parabola ($1/s^3$)



$$\frac{E(s)}{W(s)} = -\frac{Y(s)}{W(s)} = T_w(s)$$

$$T_w(s) = s^n T_{o,w}(s)$$

If **n = k**

- The system is **type k**
- The error is **1/Kn,w**.

$$y_{ss} = \lim_{s\to 0}\left[sT_w(s)\frac{1}{s^{k+1}}\right]$$

$$= \lim_{s\to 0}\left[T_{o,w}(s)\frac{s^n}{s^k}\right].$$

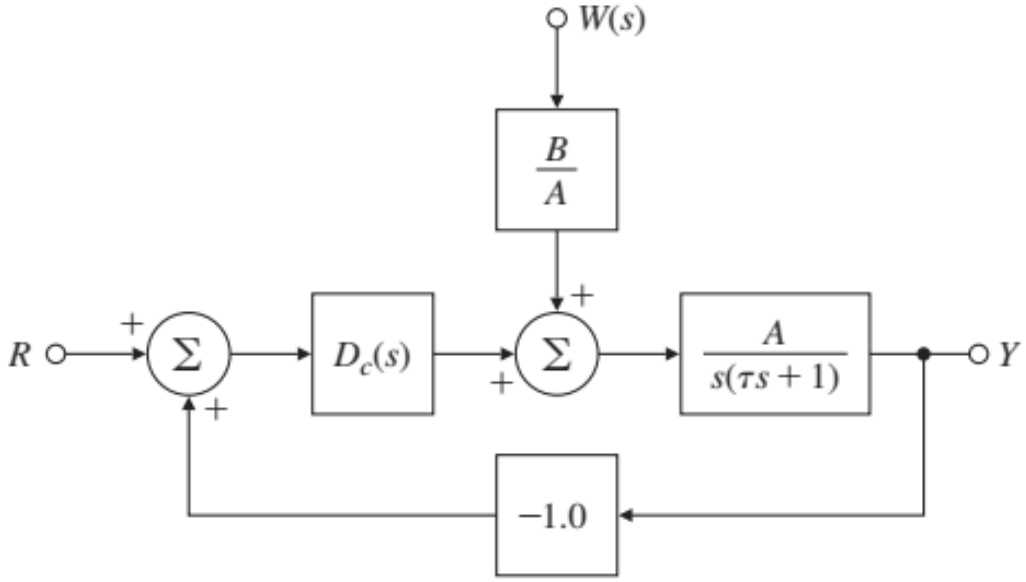
$$T_{o,w}(0) = 1/K_{n,w}$$

2. Control steady-state error to polynomial inputs: system type

Example: System Type for a DC Motor Position Control

(a) Use the controller Dc(s) = k_P and determine the system type and steady-state error properties with respect to disturbance inputs.

(b) Let the controller transfer function be given by Dc(s) = k_P + k_I/s and determine the system type and the steady-state error properties for disturbance inputs.



2. Control steady-state error to polynomial inputs: system type

Example: System Type for a DC Motor Position Control

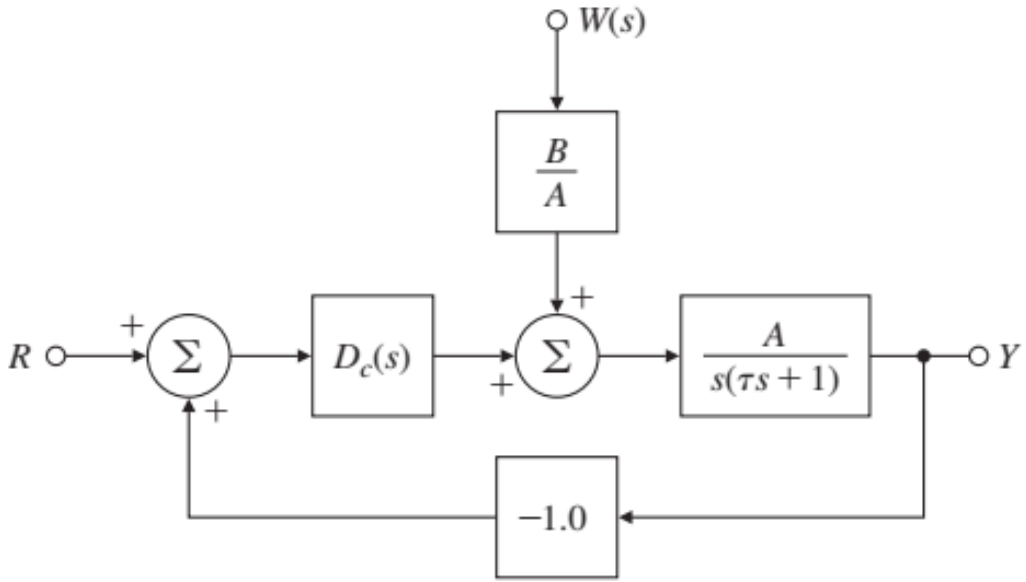
a) Dc(s) = k_P

$$T_w(s) = -\frac{B}{s(\tau s + 1) + Ak_P}$$

$$= s^0 T_{o,w},$$

$$n = 0,$$

$$K_{o,w} = -\frac{Ak_P}{B}. \qquad e_{ss} = -B/Ak_P.$$



The system is **Type 0** regarding W(s)

2. Control steady-state error to polynomial inputs: system type

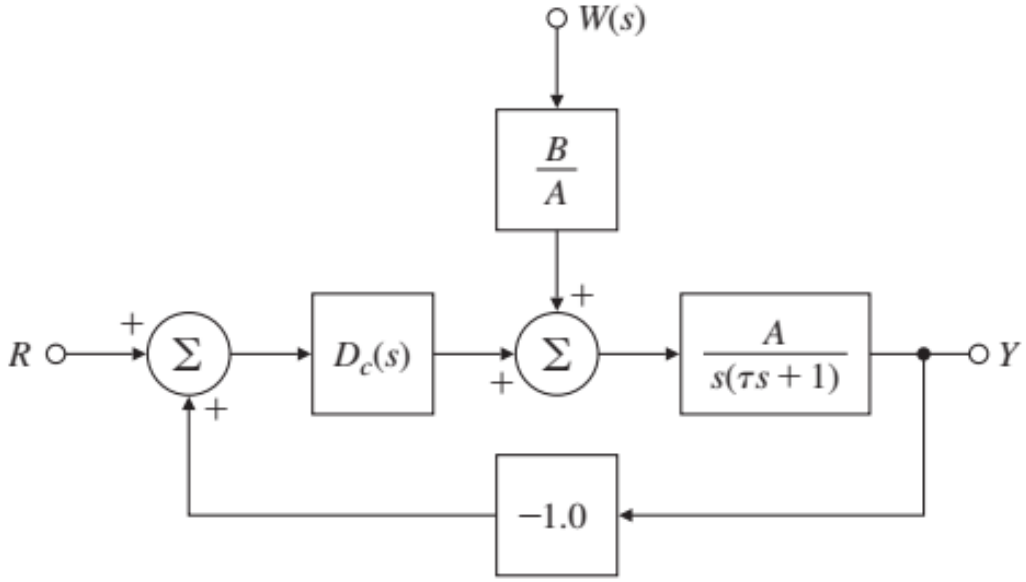
Example: System Type for a DC Motor Position Control

b) $Dc(s) = k_P + k_I/s$

$$T_w(s) = -\frac{Bs}{s^2(\tau s + 1) + (k_P s + k_I)A}$$

$$n = 1,$$

$$K_{n,w} = -\frac{Ak_I}{B}, \qquad e_{ss} = -\frac{B}{Ak_I}$$



The system is **Type 1** regarding W(s)